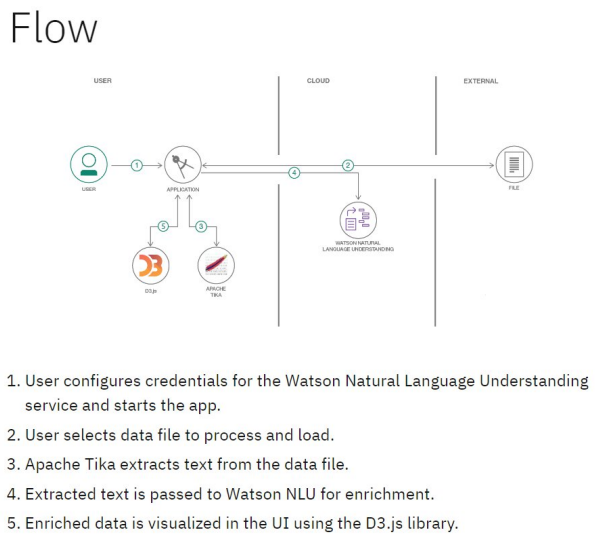
Project Design Phase-II
Data Flow Diagram & User Stories

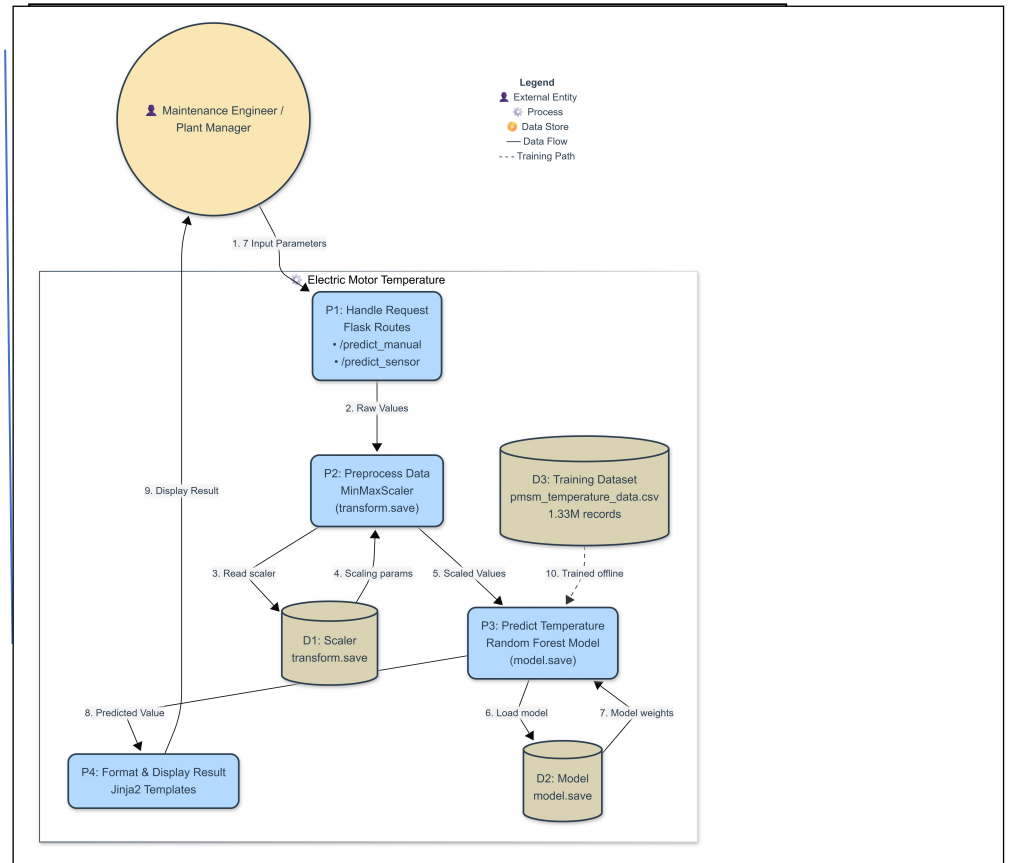
Date	12 February 2026
Team ID	LTVIP2026TMIDS81934
Project Name	Electric Motor Temperature Prediction using Machine Learning
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

Example: [(Simplified)]





User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Maintenance Engineer / Plant Manager	Temperature Prediction	USN-1	As a maintenance engineer, I want to enter 7 motor parameters (ambient, coolant, u_d, u_q, motor_speed, i_d, i_q) manually so that I can get a temperature prediction for a specific motor.	• Form displays all 7 fields with labels • Each field accepts numeric input • Submit button triggers prediction • Result displays clearly below form	High	Sprint-2
Maintenance Engineer / Plant Manager	Temperature Prediction	USN-2	As a plant manager, I want to see the predicted temperature result immediately after submitting values so that I can make quick maintenance decisions.	• Prediction appears within 2 seconds • Result shows "Predicted PM temperature: X.XX" format • Page does not refresh completely	High	Sprint-2
Maintenance Engineer / Plant Manager	Sensor Simulation	USN-3	As a maintenance engineer, I want to see auto-generated sensor values so that I can test the system without real sensor data.	• Page displays "LIVE SIMULATION" badge • All 7 fields show random values • Values displayed in card layout • "Read Sensors & Predict" button available	High	Sprint-2

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Maintenance Engineer / Plant Manager	Sensor Simulation	USN-4	As a maintenance engineer, I want to click a button to refresh sensor values and get a new prediction so that I can simulate real-time monitoring.	• Clicking button generates new random values • New prediction displays automatically • Previous result is replaced	High	Sprint-2
Maintenance Engineer / Plant Manager	Navigation	USN-5	As a user, I want a home page with clear navigation so that I can easily access manual input and sensor simulation pages.	• Home page has project title • Two buttons: "Manual Input" (green) and "Sensor Input" (blue) • Feature badges display key information	High	Sprint-2
Maintenance Engineer / Plant Manager	Model Accuracy	USN-6	As a plant manager, I want confidence that predictions are accurate so that I can trust the system for maintenance decisions.	• Model achieves 99.17% accuracy (R^2 score) • RMSE of 1.73 indicates close predictions • Accuracy badge displayed on home page	Medium	Sprint-2
System Administrator	Model Management	USN-7	As an administrator, I want the trained model and scaler to be saved and loaded automatically so that the application works without retraining.	• model.save loads at startup • transform.save loads at startup • No errors on application launch	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
System Administrator	Error Handling	USN-8	As a system administrator, I want the application to handle invalid inputs gracefully so that it doesn't crash and provides helpful error messages.	• Non-numeric inputs show error message • Empty fields show validation message • Application continues running	Medium	Sprint-2